SecureDT - User Manual

System Summary

SecureDT is a Django REST Framework application that schedules tasks across edge/MEC/cloud resources using a security-aware Genetic Algorithm. A digital twin mirrors every registered device, a Random Forest model predicts risk levels, and unplanned storage/security attacks can be simulated live.

Architecture at a glance:

- Web + REST API ..... Django 5, DRF, port 8000
- Database ..... MySQL (securetdb), at 127.0.0.1:3307
- MEC agents ..... Windows Python agent (MEC_Agent.exe)
- Cloud storage ..... Google Drive simulation module
- AI ..... Random Forest + local explainer

1. Start MySQL

```
net start SecureDT_Mysql
(port 3307, database securetdb; credentials in .env)
```

2. Install dependencies

```
cd SecureDT
python -m pip install -r requirements.txt
```

3. Migrate + create the admin user

```
python manage.py migrate
python manage.py createsuperuser
```

4. Create a pairing code, then start the server

```
python manage.py shell -c "from mec_manager.models import PairingCode;
print(PairingCode.generate_new().code)"
python manage.py runserver 0.0.0.0:8000
```

5. Log in and explore

```
Open http://127.0.0.1:8000/login
Pages: Dashboard, Task Routing, MEC Infrastructure, Digital Twin,
Genetic Algorithm, AI Intelligence, Security Monitor, Attack Simulation,
Storage Attack/Defense, Datasets, Cloud Storage, Data Flow, Analytics,
Performance, Energy Monitor.
Every page reads the live REST API; refresh polls state every 5 seconds.
```